

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
sapolyakov@sapolyakov:~$  
sapolyakov@sapolyakov:~$ ls /etc/ > file.txt  
sapolyakov@sapolyakov:~$ ls >> file.txt  
sapolyakov@sapolyakov:~$ cat file.txt  
abrt  
adjtime  
aliases  
alsa  
alternatives  
anaconda  
anacrontab  
anthy-unicode.conf  
asound.conf  
at.deny  
audit  
authselect  
avahi  
bash_completion.d  
bashrc  
bindresvport.blacklist  
binfmt.d  
bluetooth  
brlapi.key  
brltty  
brltty.conf  
ceph
```

I

Рисунок 1: Запись в файл

```
sapolyakov@sapolyakov:~$  
sapolyakov@sapolyakov:~$ grep .conf file.txt > conf.txt  
sapolyakov@sapolyakov:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprintd.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf
```

Рисунок 2: Поиск расширения .conf


```
cite.bib
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage02/report/_resources/csl
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage02/report/bib/cite.bib
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage03/report/_resources/csl
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage03/report/bib/cite.bib
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage04/report/_resources/csl
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage04/report/bib/cite.bib
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/report/_resources/csl
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/report/bib/cite.bib
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/report/_resources/csl
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/report/bib/cite.bib
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_resources/csl
/home/sapolyakov/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/bib/cite.bib
/home/sapolyakov/git-extended/.git/hooks/commit-msg.sample
/home/sapolyakov/git-extended/.git/config
/home/sapolyakov/conf.txt
sapolyakov@sapolyakov:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashe.s.d/hashe.s.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
:find: '/etc/ssh/ssh_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
sapolyakov@sapolyakov:~$  
sapolyakov@sapolyakov:~$ find ~ -name "log*" > logfile &  
[1] 6608  
sapolyakov@sapolyakov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
sapolyakov@sapolyakov:~$ rm logfile  
sapolyakov@sapolyakov:~$
```

Рисунок 5: Фоновый запуск процесса

```
sapolyakov@sapolyakov:~$  
sapolyakov@sapolyakov:~$ gedit &  
[1] 6648  
sapolyakov@sapolyakov:~$ ps | grep gedit  
    6648 pts/2    00:00:00 gedit  
sapolyakov@sapolyakov:~$ kill 6648  
[1]+  Terminated                  gedit  
sapolyakov@sapolyakov:~$  
sapolyakov@sapolyakov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DE(1)                                     User Commands                               DE(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
du(1)                                User Commands                                du(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
sapolyakov@sapolyakov:~$  
sapolyakov@sapolyakov:~$ df
```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
/dev/nvme0n1p3	155186176	89369908	65149532	58%	/
devtmpfs	4011304	0	4011304	0%	/dev
tmpfs	4035676	92	4035584	1%	/dev/shm
tmpfs	1614272	1956	1612316	1%	/run
tmpfs	1024	0	1024	0%	/run/credentials/systemd-journald.service
tmpfs	4035680	1592	4034088	1%	/tmp
/dev/nvme0n1p3	155186176	89369908	65149532	58%	/home
/dev/nvme0n1p2	1992552	518800	1352512	28%	/boot
/dev/loop1	115712	115712	0	100%	/var/lib/snapd/snap/hugo/25676
/dev/loop2	49280	49280	0	100%	/var/lib/snapd/snap/snapd/25935
/dev/loop0	68480	68480	0	100%	/var/lib/snapd/snap/core24/1349
tmpfs	1024	0	1024	0%	/run/credentials/systemd-resolved.service
tmpfs	807132	180	806952	1%	/run/user/1021
/dev/sr0	2677920	2677920	0	100%	/run/media/sapolyakov/Fedora-WS-Live-43

```
sapolyakov@sapolyakov:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/51
4      ./git-extended/.git/objects/fb
4      ./git-extended/.git/objects/85
4      ./git-extended/.git/objects/5d
4      ./git-extended/.git/objects/b3
4      ./git-extended/.git/objects/ad
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
216    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
847388 .
sapolyakov@sapolyakov:~$
```

Рисунок 10: Запуск команды du


```
/home/sapolyakov/git-extended/.git/objects/7b
/home/sapolyakov/git-extended/.git/objects/62
/home/sapolyakov/git-extended/.git/objects/9f
/home/sapolyakov/git-extended/.git/objects/6f
/home/sapolyakov/git-extended/.git/objects/8a
/home/sapolyakov/git-extended/.git/objects/51
/home/sapolyakov/git-extended/.git/objects/fb
/home/sapolyakov/git-extended/.git/objects/85
/home/sapolyakov/git-extended/.git/objects/5d
/home/sapolyakov/git-extended/.git/objects/b3
/home/sapolyakov/git-extended/.git/objects/ad
/home/sapolyakov/git-extended/.git/logs
/home/sapolyakov/git-extended/.git/logs/refs
/home/sapolyakov/git-extended/.git/logs/refs/heads
/home/sapolyakov/git-extended/.git/logs/refs/remotes
/home/sapolyakov/git-extended/.git/logs/refs/remotes/origin
/home/sapolyakov/monthly
/home/sapolyakov/reports
/home/sapolyakov/reports/monthly
/home/sapolyakov/reports/monthly/monthly
/home/sapolyakov/ski.plases
/home/sapolyakov/ski.plases/equipment
/home/sapolyakov/ski.plases/plans
/home/sapolyakov/australia
/home/sapolyakov/play
/home/sapolyakov/play/games
/home/sapolyakov/play/games/play
sapolyakov@sapolyakov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.